

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN
(Course Code: ECE 1003)

Module-1:Lecture-10
Boolean Algebra

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Module-1 (Part-2)

- ❖ Logic gates and universal gates
- ❖ Tri-state logic
- ❖ Boolean algebra
- ❖ **SOP and POS minimization**
- ❖ K-maps upto 4-variables,
- ❖ Don't care logic

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 - Sum of Minterms or sum of products (SOP)
 - Product of Maxterms or product of sums (POS)
- ❖ Standard Form
 - SOP
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BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Canonical & Standard Form

BOOLEAN ALGEBRA

- ✓ The **Boolean Algebra** can be represented in two forms
 - ✓ Canonical Form
 - ✓ Standard Form

CANONICAL FORM

- ✓ A binary variable can be either in normal form or in its complement form in a Boolean expression.

- ✓ If x and y are two binary variables, the possible AND operations are

$$x'y', x'y, xy', xy \longrightarrow \text{Called as Minterms (Standard Products)}$$

Primed binary bit – 0

Unprimed binary bit -- 1

- ✓ n variable has 2^n minterms

- ✓ Similarly, the possible OR operations of two binary variables are

$$x' + y', x' + y, x + y', x + y \longrightarrow \text{Called as Maxterms (Standard Sums)}$$

Primed binary bit – 1

Unprimed binary bit -- 0

- ✓ n variable has 2^n maxterms

CANONICAL FORM

Minterms and Maxterms for Three Binary Variables

<i>x</i>	<i>y</i>	<i>z</i>	Minterms		Maxterms	
			Term	Designation	Term	Designation
0	0	0	$x'y'z'$	m_0	$x + y + z$	M_0
0	0	1	$x'y'z$	m_1	$x + y + z'$	M_1
0	1	0	$x'yz'$	m_2	$x + y' + z$	M_2
0	1	1	$x'yz$	m_3	$x + y' + z'$	M_3
1	0	0	$xy'z'$	m_4	$x' + y + z$	M_4
1	0	1	$xy'z$	m_5	$x' + y + z'$	M_5
1	1	0	xyz'	m_6	$x' + y' + z$	M_6
1	1	1	xyz	m_7	$x' + y' + z'$	M_7

CANONICAL FORM

Sum of Minterms or Sum of Products (SOP)

- ✓ A Boolean function can be expressed algebraically from a given truth table by forming a Minterm for each combination of the variables that **produces a 1 in the function** and then **taking the OR** of all those terms.
- ✓ The minterms whose sum defines the Boolean function are those which give the 1's of the function in a truth table.
- ✓ In the example, the function F produces 1 for the input combinations 0,1 and 1,0, hence the Boolean expression can be obtained as

x	y	F	Minterms
0	0	0	
0	1	1	$x'y$
1	0	1	xy'
1	1	0	

$$F = x'y + xy' = m_1 + m_2 = \Sigma (1, 2)$$

CANONICAL FORM

Sum of Minterms or Sum of Products (SOP)

✓ **Example:** Find the Boolean expression from the truth table given

$$f_1 = x'y'z + xy'z' + xyz$$

$$= m_1 + m_4 + m_7$$

$$= \sum(1, 4, 7)$$

$$f_2 = x'yz + xy'z + xyz' + xyz$$

$$= m_3 + m_5 + m_6 + m_7$$

$$= \sum(3, 5, 6, 7)$$

Functions of Three Variables

x	y	z	Function f_1	Function f_2
0	0	0	0	0
0	0	1	1 $x'y'z$	0
0	1	0	0	0
0	1	1	0	1 $x'yz$
1	0	0	1 $xy'z'$	0
1	0	1	0	1 $xy'z$
1	1	0	0	1 xyz'
1	1	1	1 xyz	1 xyz

CANONICAL FORM

Product of Maxterms or Product of Sums (POS)

- ✓ A Boolean function can be expressed algebraically from a given truth table by forming a Maxterm for each combination of the variables that **produces a 0 in the function** and then **taking the AND** of all those terms.
- ✓ The maxterms whose product defines the Boolean function are those which give the 0's of the function in a truth table.
- ✓ In the example, the function F produces 0 for the input combinations 0,0 and 1,1, hence the Boolean expression can be obtained as

x	y	F	Maxterms
0	0	0	$x+y$
0	1	1	
1	0	1	
1	1	0	$x'+y'$

$$F = (x + y)(x' + y') = M_0 \cdot M_3 = \Pi(0, 3)$$

CANONICAL FORM

Product of Maxterms or Product of Sums (POS)

✓ **Example:** Find the Boolean expression from the truth table given

$$f_1 = (x + y + z)(x + y' + z)(x + y' + z')(x' + y$$

$$(x' + y' + z)$$

$$= M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$$

$$= \Pi (0,2,3,5,6)$$

$$f_2 = (x + y + z)(x + y + z')(x + y' + z)(x' + y$$

$$= M_0 M_1 M_2 M_4$$

$$= \Pi (0,1,2,4)$$

Functions of Three Variables

x	y	z	Function f_1	Function f_2
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

CANONICAL FORM

Example: Express the following functions in terms of sum of minterms and product of maxterms.

$$F = A + B'C$$

✓ **Sum of minterms:** Number of variable present = 3

$$F = A + B'C$$

$$= A(B + B')(C + C') + (A + A')B'C$$

$$= (AB + AB')(C + C') + AB'C + A'B'C$$

$$= ABC + ABC' + AB'C + AB'C' + A'B'C$$

$$= ABC + ABC' + AB'C + AB'C' + A'B'C$$

$$= m_7 + m_6 + m_5 + m_4 + m_1 = \Sigma(1, 4, 5, 6, 7)$$

CANONICAL FORM

Example: Express the following functions in terms of sum of minterms and product of maxterms.

$$F = A + B'C$$

✓ **Sum of maxterms:** Number of variable present = 3

$$F = A + B'C$$

$$= (A + B')(A + C)$$

$$= (A + B' + CC')(A + C + BB')$$

$$= (x + CC')(y + BB') \quad \because x = A + B' \text{ \& } y = A + C$$

$$= (x + C)(x + C')(y + B)(y + B')$$

$$= (A + B' + C)(A + B' + C')(A + B + C)(A + B' + C)$$

$$= (A + B' + C)(A + B' + C')(A + B + C) = M_2 M_3 M_0 = \Pi(0, 2, 3)$$

CANONICAL FORM

Express the Boolean function $F = xy + x'z$ as a product of maxterms. First, convert the function into OR terms by using the distributive law:

$$\begin{aligned} F &= xy + x'z = (xy + x')(xy + z) \\ &= (x + x')(y + x')(x + z)(y + z) \\ &= (x' + y)(x + z)(y + z) \end{aligned}$$

The function has three variables: x , y , and z . Each OR term is missing one variable; therefore,

$$\begin{aligned} x' + y &= x' + y + zz' = (x' + y + z)(x' + y + z') \\ x + z &= x + z + yy' = (x + y + z)(x + y' + z) \\ y + z &= y + z + xx' = (x + y + z)(x' + y + z) \end{aligned}$$

Combining all the terms and removing those which appear more than once, we finally obtain

$$\begin{aligned} F &= (x + y + z)(x + y' + z)(x' + y + z)(x' + y + z') \\ &= M_0 M_2 M_4 M_5 = \Pi(0, 2, 4, 5) \end{aligned}$$

CONVERSION BETWEEN CANONICAL FORMS

- ✓ The complement of a function expressed as the sum of minterms equals the sum of minterms missing from the original function.
- ✓ This is because the original function is expressed by those minterms which make the function equal to 1, whereas its complement is a 1 for those minterms for which the function is a 0.

$$F(A, B, C) = \Sigma(1, 4, 5, 6, 7)$$

- ✓ This function has a complement that can be expressed as

$$F'(A, B, C) = \Sigma(0, 2, 3) = m_0 + m_2 + m_3$$

- ✓ Now, if we take the complement of F by DeMorgan's theorem, we obtain F in a different form:

$$F = (m_0 + m_2 + m_3)' = m_0' m_2' m_3' = M_0 M_2 M_3 = \Pi(0, 2, 3)$$

- ✓ From the table (slide 8), the following relation holds:

$$m_j' = M_j$$

CANONICAL FORM

Example: Express the following functions in terms of sum of minterms and product of maxterms.

$$F(A, B, C, D) = B'D + A'D + BD$$

✓ **Sum of minterms:** Number of variable present = 4

$$F = B'D + A'D + BD$$

$$= (A + A')B'(C + C')D + A'(B + B')(C + C')D + (A + A')B(C + C')D$$

$$= (AB' + A'B')(CD + C'D) + (A'B + A'B')(CD + C'D) + (AB + A'B)(CD + C'D)$$

$$= AB'CD + AB'C'D + A'B'CD + A'B'C'D + A'BCD + A'BC'D + A'B'CD + A'B'C'D \\ + ABCD + ABC'D + A'B'CD + A'B'C'D$$

$$= AB'CD + AB'C'D + A'B'CD + A'B'C'D + A'BCD + A'BC'D + ABCD + ABC'D$$

$$= m_{11} + m_9 + m_3 + m_1 + m_7 + m_5 + m_{15} + m_{13} = \Sigma(1, 3, 5, 7, 9, 11, 13, 15)$$

CANONICAL FORM

Example: Express the following functions in terms of sum of minterms and product of maxterms.

$$F(A, B, C, D) = B'D + A'D + BD$$

✓ **Product of maxterms:** Number of variable present = 4

$$F = \Sigma(1, 3, 5, 7, 9, 11, 13, 15)$$

$$F = \Pi(0, 2, 4, 6, 8, 10, 12, 14)$$

CANONICAL FORM

Example: Simplify the following expression using Boolean algebra

$$F(x, y, z) = \Sigma(1, 3, 5, 7)$$

Solution: $F(x, y, z) = \Sigma(1, 3, 5, 7)$

$$= x'y'z + x'yz + xy'z + xyz$$

$$= x'z(y' + y) + xz(y' + y)$$

$$= x'z + xz$$

$$= (x' + x)z = z$$

STANDARD FORM

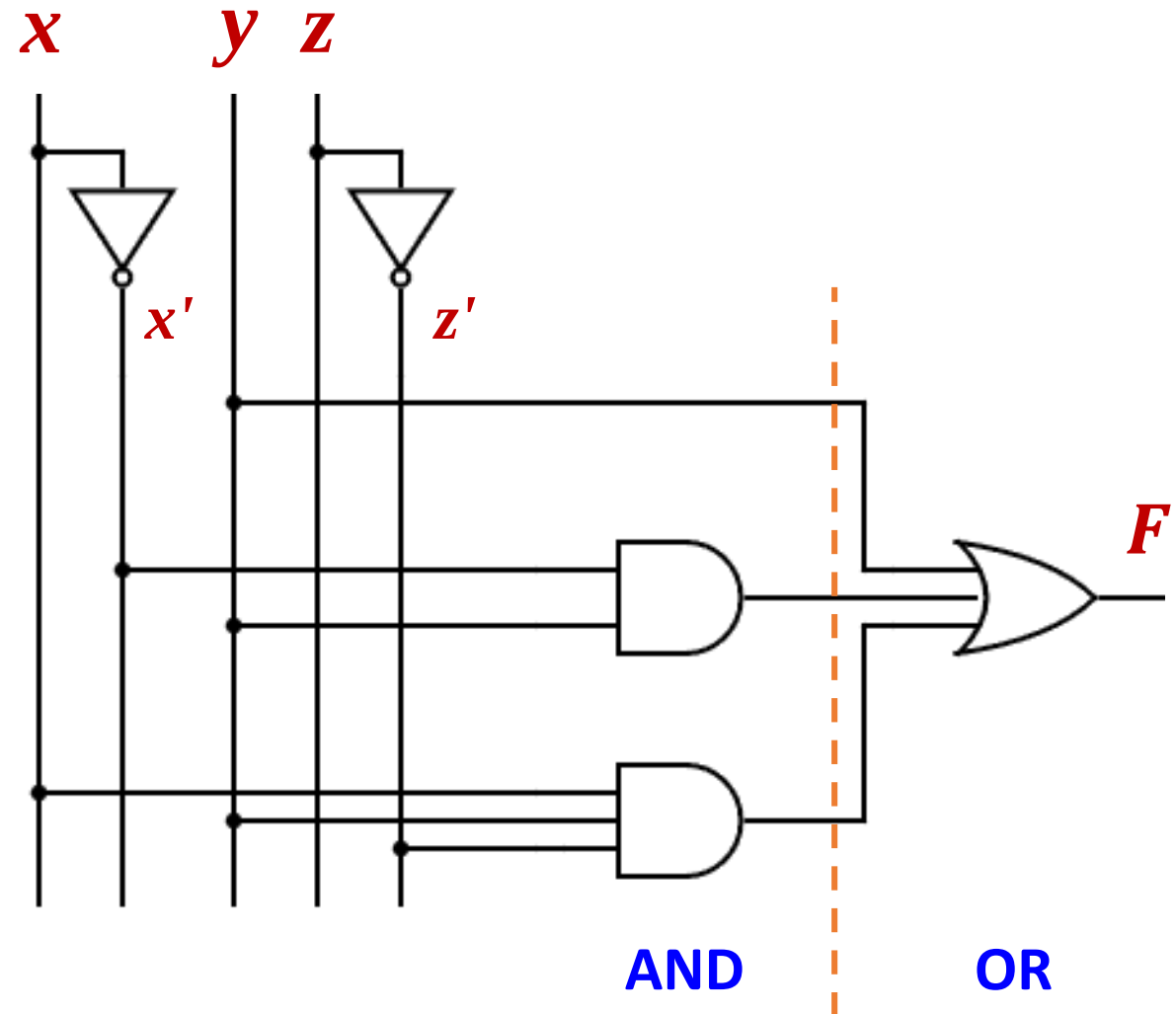
- ✓ Another way to express Boolean function is standard form representation
 - i. Sum of products (SOP)
 - ii. Product of sums (POS)

STANDARD FORM

Sum of products (SOP)

- ✓ It contains **AND terms** that may contain **one, two or any number of literals**.
- ✓ In gate level implementations, the expression consists of **a group of AND gates** followed by a **single OR gate**. Also known as **AND-OR implementation or 2-Level Implementation**.
- ✓ E.g.,

$$F = \underset{\text{2-literals}}{x'y} + \underset{\text{3-literals}}{xyz'} + \underset{\text{1-literals}}{y}$$

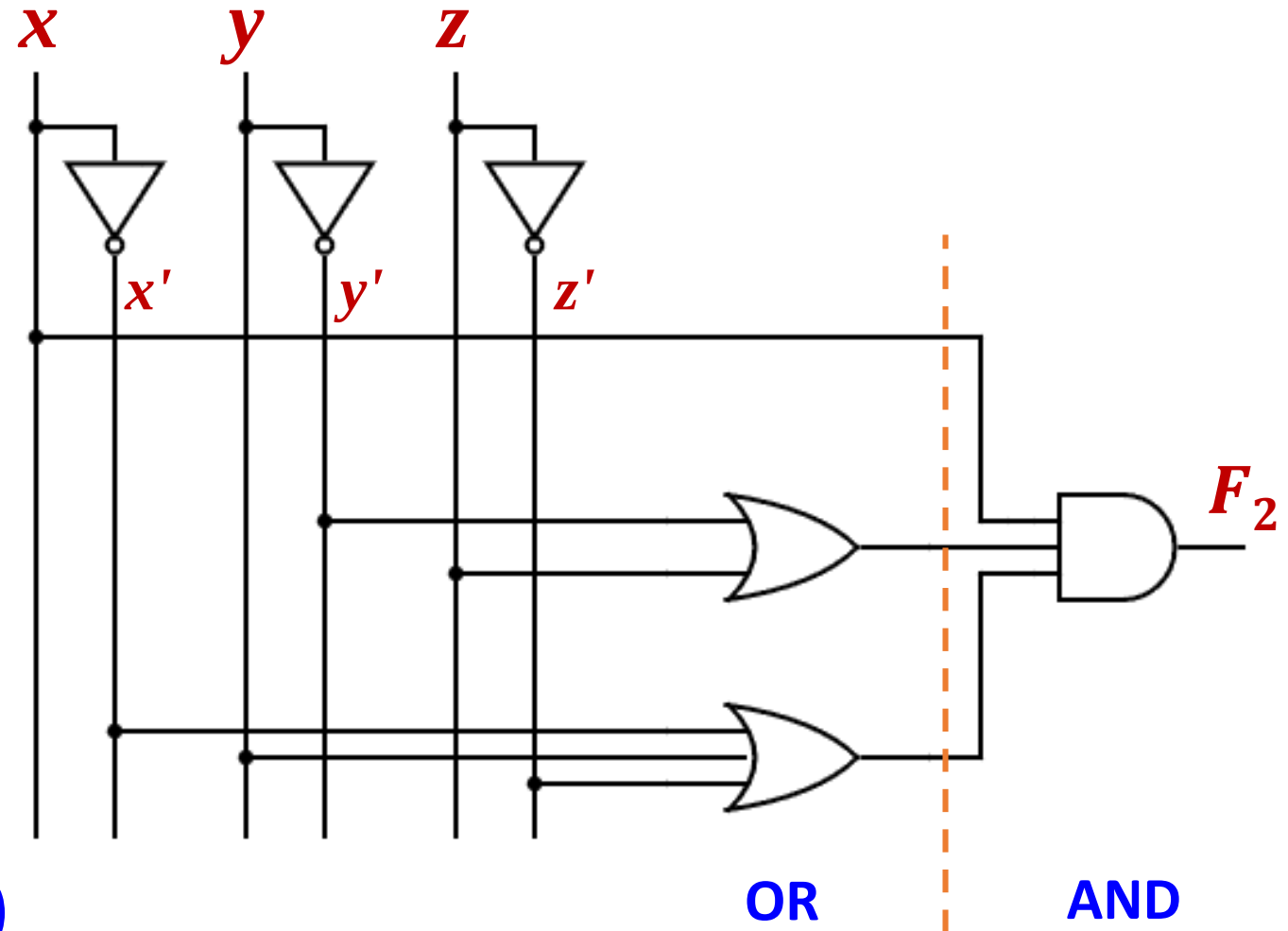


STANDARD FORM

Product of sums (POS)

- ✓ It contains **OR terms** called as sum terms.
- ✓ The product denotes the **ANDing** of these terms which contain **one, two or any number of literals**. Also known as **OR-AND implementation** or **2-Level Implementation**.
- ✓ E.g.,

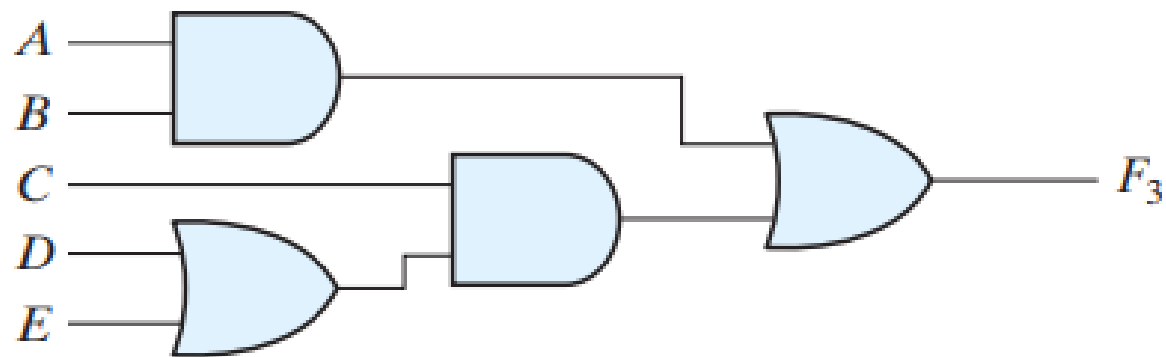
$$F_2 = x(y' + z)(x' + y + z')$$



STANDARD FORM

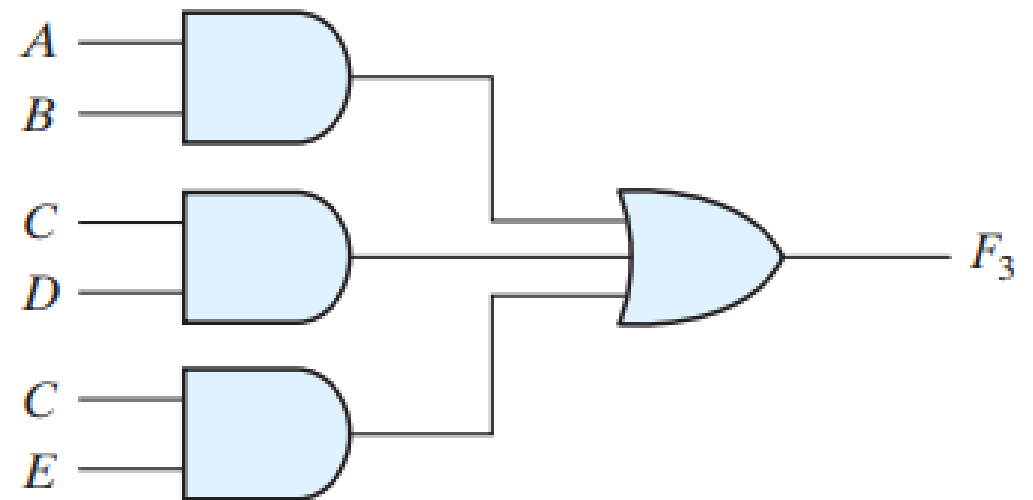
Note:

Delay is more



(a) $AB + C(D + E)$

Delay is less



(b) $AB + CD + CE$

Three- and two-level implementation

Thank You